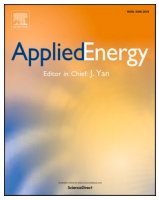
Applied Energy 401 (2025) 126800



Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy





Coordinated operation of alternative fuel vehicle-integrated microgrid in a coupled power-transportation network: a Stackelberg–Nash game framework☆

Yuyang Wan [a], Ning Wang [b,*], Ershun Du [c,*], Xueshan Liu [d], Yanbo Wang [b], Zhe Chen [b], Chongqing Kang [a]

[a] Department of Electrical Engineering, Tsinghua University, Beijing, China
[b] Department of Energy Technology, Aalborg Univercity, Aalborg 9220, Denmark
[c] Laboratory of Low Carbon Energy, Tsinghua University, Beijing 100084, China
[d] College of Electrical Engineering, Sichuan University, Sichuan 610065, China

HIGHLIGHTS

- A Stackelberg-Nash game framework is introduced to enhance the economic efficiency of AFV-integrated microgrid.
- An optimal scheduling model is formulated that accounts for the spatiotemporal dynamics and user preferences of EVs and HVs.
- A pricing and incentive mechanism is devised to promote the participation of EVs and HVs in V2G services.

ARTICLE INFO

Keywords:
Renewable energy sources
Electric vehicles
Hydrogen vehicles
Coupled power-transportation network
Stackelberg–Nash game
Vehicle-to-grid

ABSTRACT

With the rapid development of alternative fuel vehicles (AFVs) and renewable energy sources, the increasing coordination between electric vehicles (EVs) and hydrogen vehicles (HVs) in urban coupled power-transportation networks (CPTNs) fosters optimized energy scheduling and enhanced system performance. This study proposes a two-level Stackelberg-Nash game framework for AFV-integrated microgrids in a CPTN to enhance the economic efficiency of microgrid. This framework employs a Stackelberg game model to define the leader-follower relationship between the microgrid operator and the vehicle-to-grid (V2G) aggregator. Nash equilibrium games are established to capture competitive interactions among charging stations (CSs) and among hydrogen refueling stations (HRSs). Furthermore, an optimal scheduling model is proposed to minimize microgrid operation costs considering the spatiotemporal dynamics and user preferences of EVs and HVs, supported by the proposed dynamic choice model. A game-theoretic pricing and incentive mechanism promotes AFV participation in V2G services, enhancing the profitability of CSs and HRSs. Afterward, a momentum-enhanced Stackelberg-Nash equilibrium algorithm is developed to address the bi-level optimization problem. Finally, numerical simulations validate the effectiveness of the proposed method in improving economic efficiency and reducing operation costs. The proposed approach offers an effective solution for integrating large-scale AFV fleets into sustainable urban energy and transportation systems.

1. Introduction

The construction of low-carbon smart cities has emerged as a pivotal strategy in the global endeavor to address climate change and mitigate environmental degradation [1–3]. Achieving this vision depends significantly on the extensive integration of renewable energy sources (RESs) and alternative fuel vehicles (AFVs), such as electric vehicles (EVs) and hydrogen vehicles (HVs), to facilitate the green transformation of urban ecosystems [4,5]. The adoption of RESs offers a clean substitute for traditional energy frameworks, while the widespread use

☆ This article is part of a Special issue entitled: 'Hydrogen and Smart Cities(Dr. Ren)' published in Applied Energy.

* Corresponding authors.
E-mail addresses: nwa@et.aau.dk (N. Wang), duershun@tsinghua.edu.cn (E. Du).

https://doi.org/10.1016/j.apenergy.2025.126800
Received 24 July 2025; Received in revised form 12 September 2025; Accepted 17 September 2025
Available online 26 September 2025
0306-2619/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

of AFVs provides an essential avenue for decarbonizing the transportation sector [6,7]. Collectively, these technologies can enhance the operation efficiency of power systems and significantly reduce carbon emissions [8,9].

More specifically, EVs leverage vehicle-to-grid (V2G) technology to serve as mobile energy storage units, enabling them to deliver load response and frequency regulation services to the power grid [10,11]. This capability effectively addresses the fluctuations arising from the intermittent nature of RESs [12]. Simultaneously, HVs utilize fuel cell technology to support zero-emission transportation while also offering innovative approaches to renewable energy storage through hydrogen production systems [13]. Therefore, the coordinated operation of EVs and HVs has become a research hotspot [14]. The synergistic potential of these technologies by optimizing management can be fully exploited, thus facilitating a deeper integration of energy and transportation systems [15,16].

V2G technology has emerged as a transformative approach that takes the function of EVs beyond mere transportation, enabling them to act as mobile energy storage devices that feed power back into the grid, helping to stabilize electricity supply and reduce peak demand [17]. Over the past decade, a significant amount of research on V2G has focused on modeling of V2G systems [18,19], energy management of EVs [20,21], planning of charging facility [22], pricing and incentive mechanism [23,24]. For example, a state space model is proposed in [25], enabling precise and efficient real-time power control for aggregated EVs with low communication demands. A model-free hierarchical multi-agent reinforcement learning approach is proposed in [26] to optimize the routing and scheduling of EVs in a coupled power-transportation network (CPTN), effectively enhancing the provision of multiple services such as charging, demand management, carbon intensity, and grid balancing. A two-stage control framework integrating retired EV batteries in a microgrid is introduced in [27] to enhance economic efficiency through grid frequency regulation and peak shaving. A bi-level game framework modeling competitive behaviors in a CPTN operation is developed in [28] to minimize costs, reduce carbon emissions, and optimize individual entity consumption. A tri-level demand response framework, utilizing deep reinforcement learning to design incentive mechanisms for charging stations (CSs) within a CPTN, is introduced in [29] to enhance flexibility and operation efficiency while ensuring information security and independent stakeholder strategies. A comprehensive review of V2G research trends and collaborations is conducted in [30], analyzing 16,457 articles to identify gaps and suggest future directions for EV integration. These studies have shown that V2G has a promising application in supporting RESs integration and smart grid development.

Meanwhile, HVs are developing rapidly as the demand for zero-emission transportation continues to grow [38]. HVs can utilize fuel cell technology to enable emission-free mobility, and the combination of hydrogen production with RESs increases the flexibility of the smart grid [31]. In recent years, research on integrated management for electric-hydrogen systems has begun to emerge [32]. A multi-objective optimization model integrating hydrogen systems and EVs with V2G capabilities is developed in [33] to enhance RES hosting capacity by up to 17.24 % while reducing costs through coordinated management of demand response and control devices. A techno-economic analysis demonstrates that integrating EVs and hydrogen systems in an isolated grid with 75 % renewables can reduce storage costs and balance the grid [34]. An investigation into V2G technology utilizing EVs and HVs with reversible fuel cells for frequency control in an autonomous three-phase microgrid is conducted in [35]. A control framework in reconfigurable microgrids with high penetration of green smart homes, CSs, and hydrogen refueling stations (HRSs) is proposed in [36] to evaluate the impact of smart consumers and prosumers on decarbonization and local distribution system performance. A bi-level Stackelberg game-based optimization model is introduced in [37] to facilitate coordinated management of EVs and HVs, minimizing costs for the vehicle aggregator while maximizing economic benefits for power and hydrogen system operators. These studies highlight the potential of EVs and HVs to enhance grid flexibility, reduce costs, and support decarbonization; however, few have explored their coordinated operation in a CPTN.

In urban environments, the interactions between power systems and transportation networks are becoming increasingly complex [39]. The proliferation of EVs and HVs has resulted in a high degree of coupling between energy demand and vehicle usage patterns, and this coupling has a direct impact on the stability of power grids and the efficiency of transportation infrastructure [14]. Therefore, CPTN framework is critical to the study of the coordinated management of EVs and HVs [40]. Specifically, charging and refueling infrastructure can be strategically placed and managed in the CPTN to improve grid reliability, reduce operating costs, and accelerate decarbonization through integrated energy management. For example, a cooperative planning strategy optimizing RES integration in power grid and the deployment of HVs alongside EVs in transportation networks is proposed in [41] to achieve emission reductions tailored to regional power generation mixes. A multi-stage planning model for electricity-hydrogen system in a CPTN is presented in [42], addressing growing load demands and reducing carbon emissions while adapting to future policy and technological changes. A spatiotemporal model integrating EVs and HVs is developed in [43] to assess their impact and optimize costs across electricity, hydrogen, transportation, and natural gas systems.

A comparative analysis of existing and proposed methods is presented in Table 1. Despite significant progress in EVs, HVs, and V2G technologies in recent years, there are still several key challenges that have not been adequately addressed in the existing literature. In particular, there is a lack of a comprehensive framework that can effectively coordinate the management of EVs and HVs in CPTNs. Specific issues include the following: (1) Existing studies of electric-hydrogen systems typically focus only on the power system, failing to capture the complex interactions between transportation and power networks. (2) Many studies do not adequately consider the spatiotemporal dynamics of vehicle use or user choice preferences, which are critical for optimizing urban energy allocation and infrastructure utilization. (3) In AFV integrated microgrids, the competing or cooperating relationships of the various components have not been explicitly modeled, and few studies have integrated pricing strategies and operational scheduling into a unified framework. The solution of the above problems is crucial for realizing the coordinated management of EVs and HVs. Only through the comprehensive consideration of inter-system interactions, user behavior, and economic factors can an efficient and cost-effective coordinated management strategy be designed to promote the overall development of a low-carbon smart city.

Therefore, this paper proposes a two-level management framework for the AFV-integrated microgrid, designed to tackle the dual challenges

Table 1
A comparative analysis of existing and proposed methods.

Ref.	EV	HV	Hydrogen system	CPTN	Pricing strategy	EV user behavior
[26]	Yes	No	No	Yes	No	Yes
[27]	Yes	No	No	No	No	Yes
[28]	Yes	No	No	Yes	Yes	Yes
[29]	Yes	No	No	Yes	Yes	No
[31]	Yes	Yes	No	No	No	No
[32]	No	Yes	Yes	No	Yes	No
[33]	Yes	No	Yes	No	No	No
[34]	Yes	No	Yes	No	No	No
[35]	Yes	Yes	No	No	No	No
[36]	Yes	Yes	Yes	No	Yes	Yes
[37]	Yes	Yes	Yes	No	Yes	No
This paper	Yes	Yes	Yes	Yes	Yes	Yes

of power scheduling and pricing mechanisms. The key contributions of this work are outlined below:

(1) A Stackelberg-Nash game framework is introduced to enhance the economic efficiency of microgrid. In this framework, the leader-follower relationship between the microgrid operator and the V2G aggregator is defined, with Nash equilibrium games established to model the competitive interactions among CSs and among HRSs. A momentum-enhanced Stackelberg-Nash equilibrium algorithm (MES-NEA) is developed to address the bi-level optimization problem inherent in the proposed framework.

(2) An optimal scheduling model is formulated that accounts for the spatiotemporal dynamics and user preferences of EVs and HVs, with the objective of minimizing microgrid operation costs. Furthermore, a dynamic choice model is introduced to capture user preferences for various stations in a CPTN, incorporating realistic factors such as pricing, geographic proximity, vehicle status, and service quality.

(3) A pricing and incentive mechanism, grounded in non-cooperative game theory, is devised to promote the participation of EVs and HVs in V2G services. By leveraging competitive pricing and subsidies, this mechanism enhances the overall profitability of both charging and hydrogen refueling stations.

It is worth noting that the proposed framework still faces several limitations. First, the model only captures unidirectional coupling relationships while overlooking the potential for concentrated charging activities to trigger local traffic congestion. Second, it assumes perfect information sharing between the microgrid operator and V2G aggregator, which can be challenging given market competition and privacy concerns. These limitations motivate future research in two directions: (1) Developing bidirectional dynamic coupling models that integrate traffic simulation with energy optimization to capture full feedback effects. (2) Designing distributed optimization frameworks with privacy-preserving mechanisms for efficient coordination without complete information transparency.

The rest of this paper is organized as follows. In Section 2, the system description is given. In Section 3, the problem formulation is introduced. In Section 4, the Stackelberg–Nash hierarchical game framework is proposed. The simulation results are provided to validate the theoretical analysis in Section 5. Finally, some conclusions are given in Section 6.

2. System description

This paper proposes a novel configuration of an AFV-integrated microgrid in an urban CPTN to enhance operation efficiency and optimize energy scheduling. As illustrated in Fig. 1, the microgrid integrates distributed generation units, including photovoltaic panels (PV) and wind turbines (WT), alongside CSs and HRSs with hydrogen production unit, serving industrial and residential loads. The microgrid control center coordinates these components, maintaining bidirectional connectivity with the main grid to enable energy exchange and support demand response and ancillary services.

To manage the interdependencies among EVs, HVs, the power system, and the transportation network, a two-level management framework is developed, integrating centralized energy scheduling with decentralized station and user behavior modeling. The upper-level optimization in microgrid control center can minimize the total operation costs of microgrid. Formulated as a mixed-integer linear programming (MILP) problem, this optimization adheres to power balance and capacity constraints across the loads of microgrid. The expected CS and HRS power capacities are derived from the lower-level model, reflecting user participation and station operations. At the lower level, a non-cooperative game-theoretic model captures the competitive interactions among multiple stations. Each station optimizes its pricing strategy, comprising charging prices, V2G discharging subsidies, and hydrogen prices, to maximize profits, considering the response behavior of EV and HV users. A logit-based choice model estimates user selection probabilities, accounting for pricing, geographic proximity, vehicle status, and service quality. The resulting Nash equilibrium determines the anticipated vehicle participation and available V2G and hydrogen resources, which are fed back to the upper-level optimization.

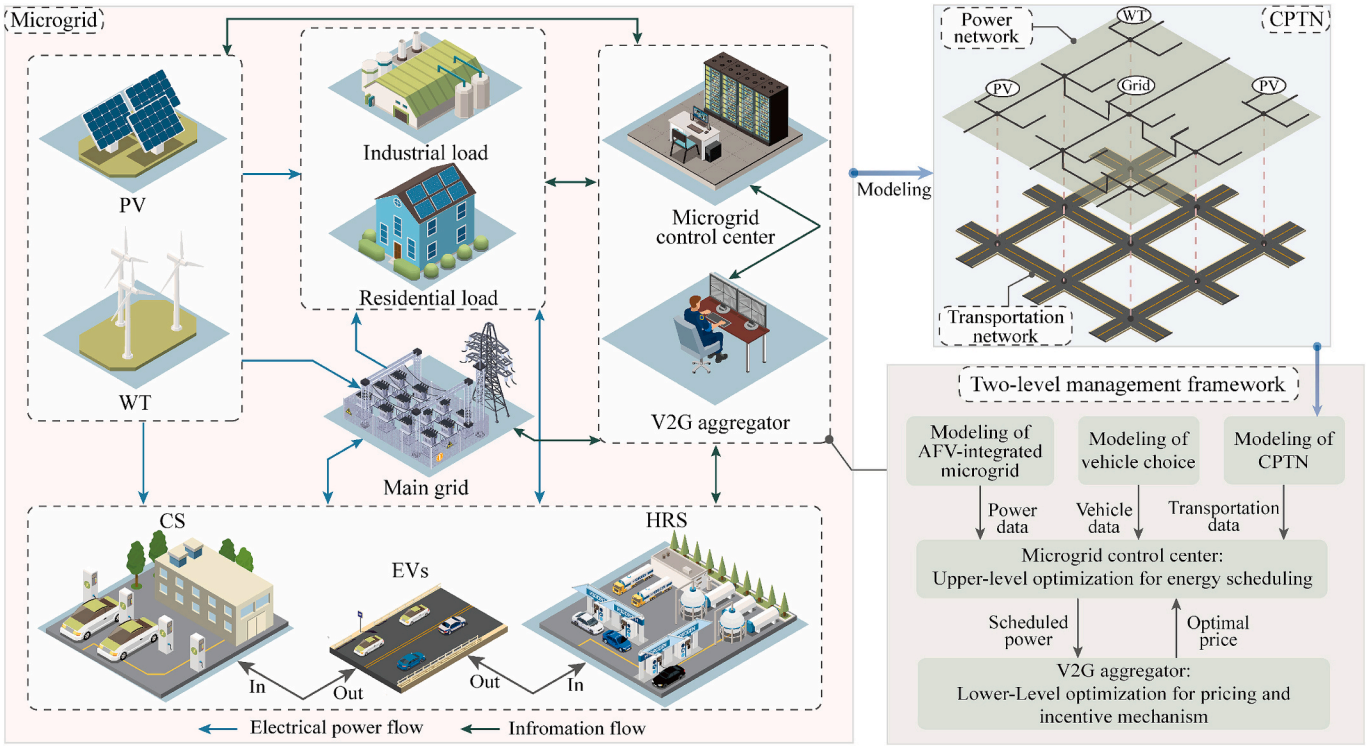


Fig. 1. The proposed AFV-integrated microgrid and its management framework.

3. Problem formulation

This section presents the optimization problem formulation for coordinating the management of an AFV-integrated microgrid in a CPTN. The formulation is organized into objective functions, dynamic choice model of EVs and HVs in CPTN, and the associated constraints.

3.1. Objective functions

3.1.1. Scheduling strategy of the proposed microgrid

The primary goal of the microgrid scheduling strategy is to minimize the total operation cost over a specified time horizon, which can be mathematically expressed by (1). The operation cost of microgrid C_{MG} includes expenses related to RESs, interactions with the main grid, CS operations, and HRS operations.

$$\min C_{\mathrm{MG}} = \sum_{t=1}^{T} \left(C_{\mathrm{RES},t} + C_{\mathrm{grid},t} + C_{\mathrm{CS},t} + C_{\mathrm{HRS},t}\right) \tag{1}$$

in which:

$$C_{\mathrm{RES},t} = c_{\mathrm{PV}} \sum_{j \in N_{\mathrm{b}}} P_{\mathrm{PV},j,t} + c_{\mathrm{WT}} \sum_{j \in N_{\mathrm{b}}} P_{\mathrm{WT},j,t} \tag{2}$$

$$C_{\mathrm{grid},t} = c_{\mathrm{e},t} \sum_{j \in N_{\mathrm{b}}} P_{\mathrm{grid},j,t} \tag{3}$$

$$C_{\mathrm{CS},t} = c_{\mathrm{EV}} \sum_{m \in N_{\mathrm{EV}}} \left(P_{\mathrm{dc,EV}m,t} + P_{\mathrm{ch,EV}m,t}\right) + \sum_{j \in N_{\mathrm{b}}} \left(c_{\mathrm{V2G},j,t} P_{\mathrm{dc},j,t} - c_{\mathrm{ch},j,t} P_{\mathrm{ch},j,t}\right) \tag{4}$$

$$C_{\mathrm{HRS},t} = \sum_{j \in N_{\mathrm{b}}} \left(c_{\mathrm{HRS}} \left|P_{\mathrm{HRS},j,t}\right| + c_{\mathrm{h},t} H_{\mathrm{HRS},j,t} - c_{\mathrm{HRS},j,t} D_{\mathrm{HRS},j,t}\right) \tag{5}$$

where $C_{\mathrm{RES},t}$, $C_{\mathrm{grid},t}$, $C_{\mathrm{CS},t}$ and $C_{\mathrm{HRS},t}$ represent the operation costs of RESs, main grid, CSs and HRSs at time t, c_{PV}, c_{WT}, c_{CS} and c_{HRS} stand for the unit operation costs of PV, WT, CS and HRS, $c_{\mathrm{e},t}$ is the electricity price at time t, $P_{\mathrm{PV},j,t}$, $P_{\mathrm{WT},j,t}$, $P_{\mathrm{grid},j,t}$ and $P_{\mathrm{HRS},j,t}$ represent the operation power of PV, WT, main grid and HRS at bus j and time t, $P_{\mathrm{dc,EV}m,t}$ and $P_{\mathrm{ch,EV}m,t}$ represent the discharging and charging power of m-th EV at time t.

3.1.2. Incentive mechanism of V2G aggregator

The objective of the proposed incentive mechanism focuses on maximizing the profit of each CS through strategic pricing. The revenue of CS and HRS at bus j can be obtained as:

$$F_{\mathrm{CS}j} = \sum_{t \in T} \left[\left(c_{\mathrm{ch},j,t} - c_{\mathrm{e},t}\right) P_{\mathrm{ch},j,t} - \left(c_{\mathrm{V2G},j,t} - c_{\mathrm{e},t}\right) P_{\mathrm{dc},j,t}\right], \forall j \in N_{\mathrm{b}} \tag{6}$$

$$F_{\mathrm{HRS},j} = \sum_{t \in T} \left(c_{\mathrm{HRS},j,t} D_{\mathrm{HRS},j,t} - c_{\mathrm{h},t} H_{\mathrm{HRS},j,t} - c_{\mathrm{e},t} P_{\mathrm{HRS},j,t}\right), \forall j \in N_{\mathrm{b}} \tag{7}$$

where $c_{\mathrm{ch},j,t}$ and $c_{\mathrm{V2G},j,t}$ are the charging price and V2G of CS at bus j and time t, $P_{\mathrm{dc,j},t}$ and $P_{\mathrm{ch,j},t}$ represent the discharging and charging power at bus j and time t, $D_{\mathrm{HRS},j,t}$ is the hydrogen demand of HRS at bus j and time t, $H_{\mathrm{HRS},j,t}$ is the hydrogen purchase of HRS at bus j and time t, $c_{\mathrm{HRS}i,t}$ is the hydrogen sale price of HRS at bus j and time t, $c_{\mathrm{h},t}$ is the hydrogen purchase price at time t.

3.2. Dynamic choice model of EVs and HVs in CPTN

The CPTN can be effectively represented using a graph-based model, denoted as $G = (N, E)$. In this framework, the set N represent transportation nodes and edges E define the feasible travel paths linking these locations. $N_{\mathrm{b}} = (1, 2, 3, \ldots, n_{\mathrm{b,max}})$ represents the set of all buses of the grid, and L_{b} denotes the set of all transmission lines. A critical feature of this model is the mapping of buses in the power grid to specific nodes in the transportation network. The distance matrix D contains the shortest path distances $d_{i,j}$ between nodes, calculated using the Floyd-Warshall algorithm.

$$\begin{cases} N = \{1, 2, 3, \ldots, n_{\max}\} \\ E = \left\{e_{\mathrm{i,j}}\right\}, \forall i \neq j \in N \\ D = \left[d_{ij}\right], \forall (i,j) \end{cases} \tag{8}$$

This model captures the dynamic relationship between traffic conditions and energy infrastructure by linking key components. Under congested conditions, the traffic flow $\rho_{ij,t}$ can increase travel time according to a Bureau-of-Public-Roads–type function:

$$t_{ij}\left(\rho_{ij,t}\right) = t_{ij}^{0}\left[1 + 0.15\left(\frac{\rho_{ij,t}}{c_{ij}}\right)^{4}\right] \tag{9}$$

where c_{ij} is the capacity of edge $e_{\mathrm{i,j}}$. For any simple path $P_{rs} \in \mathcal{P}$ connecting an origin-destination (O—D) pair $rs \in D_{\mathrm{rs}}$, the total travel time at time t is the sum of its edge travel times:

$$T_{P_{rs},t} = \sum_{(i,j) \in P_{rs}} t_{ij}\left(\rho_{ij,t}\right) \tag{10}$$

The minimal travel time between an O—D pair $rs \in D_{\mathrm{rs}}$ at time t is obtained as:

$$T_{rs,t} = \min_{P_{rs} \in \mathcal{P}} \sum_{(i,j) \in P_{rs}} t_{ij}\left(\rho_{ij,t}\right), \quad (r,s) \in D_{rs} \tag{11}$$

Each EV or HV user begins their trip at a specific origin and concludes it at a designated destination, forming what is referred to as an O—D pair. These O—D pairs define the spatial scope of user movement and serve as the foundation for modeling vehicle flows across the transportation network. The trip departure time for both EV and HV users, defined as the moment each vehicle begins moving from its origin toward its destination, is modeled as a stochastic variable over space and time. Each EV is further described by three characteristics: its initial state of charge (SoC) at departure, its target SoC upon completion of charging, and its expected dwell time at the charging station. In contrast, each HV user is characterized solely by the required hydrogen volume, since refueling durations are typically negligible compared with battery charging times. The network travel time from a user's origin to a selected CS or HRS can be calculated in (11).

In this work, the dynamic choice model is developed to capture the dynamic behavior of EV and HV users in selecting CS or HRS [44]. This model leverages the discrete choice model, grounded in random utility theory, to model user preferences. For both EV and HV users, utility functions are constructed in (12) and (13) to capture these decision-making processes. Each utility function consists of a deterministic component and a stochastic error term. The stochastic error term reflects unobserved user preferences or external factors assumed to follow the Gumbel distribution, which is the basis of the Logit model.

$$U_{\mathrm{EV}m,j,t} = V_{\mathrm{EV}m,j,t} + \epsilon_{\mathrm{EV}m,j}, \forall m \in N_{\mathrm{EV}} \tag{12}$$

$$U_{\mathrm{HV}m,j,t} = V_{\mathrm{HV}m,j,t} + \epsilon_{\mathrm{HV}m,j}, \forall m \in N_{\mathrm{HV}} \tag{13}$$

For EV users, deterministic utility is a weighted linear combination of four factors: price, distance, incentive, and service quality. For HV users, deterministic utility is affected by hydrogen price, distance and service quality. Price and distance typically have a negative impact on utility, while incentive and service quality have a positive impact. Service quality represents the overall user experience at charging/refueling stations, encompassing multiple dimensions including: (1) Average waiting time at the station. (2) Availability of charging/refueling points. (3) Reliability of equipment. (4) Amenities available. (5) Customer service and technical support.

$$V_{\mathrm{EV}m,j,t} = \beta_1 c_{\mathrm{ch},j,t} + \beta_2 d_{\mathrm{EV}m,j} + \beta_3 c_{\mathrm{V2G},j,t} + \beta_4 S_{\mathrm{CS},j}, \forall m \in N_{\mathrm{EV}} \tag{14}$$

$$V_{\mathrm{HV}m,j,t} = \gamma_1 c_{\mathrm{HRS},j,t} + \gamma_2 d_{\mathrm{HV}m,j} + \gamma_3 S_{\mathrm{HRS},j}, \forall m \in N_{\mathrm{HV}} \tag{15}$$

Considering the case where EV or HV does not choose any CS or HRS,

the deterministic utility function at this time can be expressed as:

$$V_{\mathrm{EV}m,0,t} = \frac{\beta_1}{n_{\mathrm{CS}}} \sum_{j \in N_{\mathrm{b}}} c_{\mathrm{ch},j,t} + \beta_5 \mathrm{SoC}_{\mathrm{EV}m,t}, \forall m \in N_{\mathrm{EV}} \tag{16}$$

$$V_{\mathrm{HV}m,0,t} = \frac{\gamma_1}{n_{\mathrm{HRS}}} \sum_{j \in N_{\mathrm{b}}} c_{\mathrm{HRS},j,t} + \gamma_4 \mathrm{SoC}_{\mathrm{HV}m,t}, \forall m \in N_{\mathrm{HV}} \tag{17}$$

where $d_{\mathrm{EV}m,j}$ is the distance from m-th EV to the CS at bus j, $S_{\mathrm{CS},j}$ and $S_{\mathrm{HRS},j}$ represent the quality of service, $d_{\mathrm{HV}m,j}$ is the distance from m-th HV to the HRS at bus j, $\mathrm{SoC}_{\mathrm{EV}m,t}$ and $\mathrm{SoC}_{\mathrm{HV}m,t}$ represent the SoC of m-th EV and m-th HV at time t. β_1, β_2, β_3, β_4, β_5, γ_1, γ_2, γ_3 and γ_4 are the coefficients.

Therefore, the EVs and HVs charging decisions can be formulated as follows:

$$\delta_{\mathrm{EV}m,t} = \begin{cases} 1, \mathrm{if} \max\left(V_{\mathrm{EV}m,j,t}\right) \geq V_{\mathrm{EV}m,0,t} \\ 0, \mathrm{otherwise} \end{cases}, \forall m \in N_{\mathrm{EV}} \tag{18}$$

$$\delta_{\mathrm{HV}m,t} = \begin{cases} 1, \mathrm{if} \max\left(V_{\mathrm{HV}m,j,t}\right) \geq V_{\mathrm{HV}m,0,t} \\ 0, \mathrm{otherwise} \end{cases}, \forall m \in N_{\mathrm{HV}} \tag{19}$$

The binary decision variables $\delta_{\mathrm{EV}m,t}$ and $\delta_{\mathrm{HV}m,t}$ determine whether the m-th EV or m-th HV chooses to charge or refuel at time t. For EVs, if the maximum deterministic utility $\max(V_{\mathrm{EV}m,j,t})$ across all charging stations exceeds or equals the non-charging utility $V_{\mathrm{EV}m,0,t}$, then $\delta_{\mathrm{EV}m,t} = 1$, indicating the m-th EV chooses to charge; otherwise, $\delta_{\mathrm{EV}m,t} = 0$. Similarly, eq. (19) applies the same logic to HVs.

Therefore, based on the Logit model, the choice probability of users at time t can be expressed as:

$$p_{\mathrm{EV}m,j,t} = \frac{\exp\left(V_{\mathrm{EV}m,j,t}\right)}{\sum_{j=1}^{N_{\mathrm{EV}}} \exp\left(V_{\mathrm{EV}m,j,t}\right)}, \forall m \in N_{\mathrm{EV}} \tag{20}$$

$$p_{\mathrm{HV}m,j,t} = \frac{\exp\left(V_{\mathrm{HV}m,j,t}\right)}{\sum_{j=1}^{N_{\mathrm{HV}}} \exp\left(V_{\mathrm{HV}m,j,t}\right)}, \forall m \in N_{\mathrm{HV}} \tag{21}$$

where $p_{\mathrm{EV}m,j,t}$ represents the probability that m-th EV user selects the CS at bus j, based on the relative utility of that station compared to all available options. Similarly, $p_{\mathrm{HV}m,j,t}$ represents the probability that m-th HV user selects the HRS at bus j.

3.3. Constraints

The constraints ensure the operation feasibility of the microgrid and are divided into three subsections as requested: power network constraints, CS constraints, and HRS constraints.

3.3.1. Power network constraints

The operation of the power network is defined by constraints (22)–(30) [45]. Constraints (22) and (23) ensure the balance of active and reactive power at each bus in the power network. Constraint (24) establishes a quadratic relationship between the voltage drop across line (i,j) and the corresponding power flow. Constraints (25) and (26) delineate the allowable ranges for the apparent power of each line ($S_{ij,t}$) and the squared voltage magnitude of each bus ($v_{j,t}$), respectively. Constraints (27)–(30) impose limits on the active and reactive power generation of PV and WT operating under the specified power factors $\varphi_{\mathrm{PV,j}}$ and $\varphi_{\mathrm{WT,j}}$.

$$\begin{aligned} P_{\mathrm{grid},j,t} + P_{\mathrm{PV},j,t} + P_{\mathrm{WT},j,t} + P_{ij,t} - r_{ij}\ell_{ij,t} = \sum_{k \in \Omega(j)} P_{jk,t} \\ + P_{\mathrm{load},j,t} + P_{\mathrm{CS},j,t} + P_{\mathrm{HRS},j,t}, \quad \forall j \in N_{\mathrm{b}} \end{aligned} \tag{22}$$

$$\begin{aligned} Q_{\mathrm{grid},j,t} + Q_{\mathrm{PV},j,t} + Q_{\mathrm{WT},j,t} + Q_{ij,t} - x_{ij}\ell_{ij,t} = \sum_{k \in \Omega(j)} Q_{jk,t} \\ + Q_{\mathrm{load},j,t}, \quad \forall j \in N_{\mathrm{b}} \end{aligned} \tag{23}$$

$$v_{i,t} - v_{j,t} = 2\left(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}\right) - \left(r_{ij}^2 + x_{ij}^2\right)\ell_{ij,t}, \forall (i,j) \in L_{\mathrm{b}} \tag{24}$$

$$S_{ij,t}^2 \geq P_{ij,t}^2 + Q_{ij,t}^2, \forall (i,j) \in L_{\mathrm{b}} \tag{25}$$

$$v_{j,\min} \leq v_{j,t} \leq v_{j,\max}, \forall j \in N_{\mathrm{b}} \tag{26}$$

$$0 \leq P_{\mathrm{PV},j,t} \leq P_{\mathrm{PV},j,\max}, \forall j \in N_{\mathrm{b}} \tag{27}$$

$$Q_{\mathrm{PV},j,t} \leq \tan\left[\arccos\left(\varphi_{\mathrm{PV},j}\right)\right] P_{\mathrm{PV},j,t}, \forall j \in N_{\mathrm{b}} \tag{28}$$

$$0 \leq P_{\mathrm{WT},j,t} \leq P_{\mathrm{WT},j,\max}, \forall j \in N_{\mathrm{b}} \tag{29}$$

$$Q_{\mathrm{WT},j,t} \leq \tan\left[\arccos\left(\varphi_{\mathrm{WT},j}\right)\right] P_{\mathrm{WT},j,t}, \forall j \in N_{\mathrm{b}} \tag{30}$$

where $P_{ij,t}$ and $Q_{ij,t}$ represent the active and reactive power on line (i,j), $\Omega(j)$ is the set of buses stemming from bus j, $\ell_{ij,t}$ represents the square of current magnitude for line (i,j), x_{ij} and r_{ij} denote the reactance and resistance of line (i,j).

3.3.2. CS constraints

The formulations (31)–(34) establish the operation constraints and dynamic behavior of CS. Eqs. (31) and (32) define the power distribution mechanism for charging and discharging operations respectively, where the actual power delivered to each CS at time t is calculated through binary decision variables $\delta_{\mathrm{EV}m,t}$ and choice probability. Constraint (33) enforces temporal and operational restrictions, ensuring that charging and discharging powers are zero when EVs are not connected at the station, maintaining mutual exclusivity between charging and discharging operations, and limiting power levels within predetermined bounds. Finally, eq. (34) models the SoC dynamics, incorporating upper and lower SoC limits, the update mechanism of SoC, and the target SoC requirement that must be met at the end of EV charging.

$$P_{\mathrm{ch},j,t} = \sum_{m \in N_{\mathrm{EV}}} \delta_{\mathrm{EV}m,t} p_{\mathrm{EV}m,j,t} P_{\mathrm{ch,EV}m,t}, \forall j \in N_{\mathrm{b}} \tag{31}$$

$$P_{\mathrm{dc},j,t} = \sum_{m \in N_{\mathrm{EV}}} \delta_{\mathrm{EV}m,t} p_{\mathrm{EV}m,j,t} P_{\mathrm{dc,EV}m,t}, \forall j \in N_{\mathrm{b}} \tag{32}$$

$$\begin{cases} P_{\mathrm{ch,EV}m,t} = P_{\mathrm{dc,EV}m,t} = 0, t \leq t_{\mathrm{b,EV}m} or t \geq t_{\mathrm{f,EV}m} \\ \delta_{\mathrm{ch,EV}m,t} + \delta_{\mathrm{dc,EV}m,t} \leq 1, \forall t \in T \\ 0 \leq P_{\mathrm{ch,EV}m,t} \leq P_{\mathrm{ch,EV,max}}, t_{\mathrm{b,EV}m} \leq t \leq t_{\mathrm{f,EV}m} \\ 0 \leq P_{\mathrm{dc,EV}m,t} \leq P_{\mathrm{dc,EV,max}}, t_{\mathrm{b,EV}m} \leq t \leq t_{\mathrm{f,EV}m} \end{cases}, \forall m \in N_{\mathrm{EV}} \tag{33}$$

$$\begin{cases} \mathrm{SoC}_{\mathrm{EV,min}} \leq \mathrm{SoC}_{\mathrm{EV}m,t} \leq \mathrm{SoC}_{\mathrm{EV,max}} \\ \mathrm{SoC}_{\mathrm{EV}m,t+1} = \mathrm{SoC}_{\mathrm{EV}m,t} + \dfrac{P_{\mathrm{ch,EV}m,t} - P_{\mathrm{dc,EV}m,t}}{Q_{\mathrm{EV}}} \Delta t, \forall m \in N_{\mathrm{EV}} \\ \mathrm{SoC}_{\mathrm{EV}m,t} \geq \mathrm{SoC}_{\mathrm{EV}m,\mathrm{tar}}, t = t_{\mathrm{b,EV}m} \end{cases} \tag{34}$$

where $t_{\mathrm{b,EV}m}$ and $t_{\mathrm{f,EV}m}$ are the begin time and end time of the EV's connection to the CS, respectively. $\delta_{\mathrm{ch,EV}m,t}$ and $\delta_{\mathrm{dc,EV}m,t}$ are binary variables representing whether the m-th EV is charging or discharging at time t. $P_{\mathrm{ch,EV,max}}$ and $P_{\mathrm{dc,EV,max}}$ represent the maximum charging power and maximum discharging power of EV. $\mathrm{SoC}_{\mathrm{EV}m,t}$ is SoC of m-th EV at time t, Q_{EV} is the actual capacity of EV battery, $\mathrm{SoC}_{\mathrm{EV}m,\mathrm{tar}}$ is target SoC of m-th EV, $\mathrm{SoC}_{\mathrm{EV,max}}$ and $\mathrm{SoC}_{\mathrm{EV,min}}$ denote the maximum and minimum limits of SoC for EVs.

3.3.3. HRS constraints

The formulations (35)–(38) establish the operation constraints for HRS equipped with electrolysis-based hydrogen production capabilities.

These stations can produce hydrogen through water electrolysis, store it in onsite tanks, and provide refueling services to hydrogen vehicles or grid support through fuel cell operation. Eq. (35) defines the hydrogen demand of HRS at bus j. Eq. (36) defines the power consumption of the HRS as the difference between electrolyzer power consumption $P_{\mathrm{el},j,t}$ and fuel cell power generation $P_{\mathrm{fc},j,t}$. Constraint (37) enforces the operational logic and power limits, ensuring mutual exclusivity between water electrolyzer and fuel cell operations through binary variables, while constraining both electrolyzer and fuel cell power within their respective operation bounds. When $\delta_{\mathrm{el},j,t} = 1$, the electrolyzer will work; when $\delta_{\mathrm{fc},j,t} = 1$, the fuel cell will be run. Constraint (38) models the hydrogen storage dynamics of the HRS, incorporating upper and lower limits for the SoC, the update mechanism of SoC, and the boundary condition ensuring initial and final SoC equality for operation cyclicity.

$$D_{\mathrm{HRS},j,t} = \sum_{m \in N_{\mathrm{HV}}} \delta_{\mathrm{HV}m,t} p_{\mathrm{HV}m,j,t} d_{\mathrm{HV}m}, \forall j \in N_{\mathrm{b}} \tag{35}$$

$$P_{\mathrm{HRS},j,t} = P_{\mathrm{el},j,t} - P_{\mathrm{fc},j,t}, \forall j \in N_{\mathrm{b}} \tag{36}$$

$$\begin{cases} \delta_{\mathrm{el},j,t} + \delta_{\mathrm{fc},j,t} \le 1 \\ 0 \le P_{\mathrm{el},j,t} \le \delta_{\mathrm{el},j,t} P_{\mathrm{el},j,\max} \\ 0 \le P_{\mathrm{fc},j,t} \le \delta_{\mathrm{fc},j,t} P_{\mathrm{fc},j,\max} \end{cases}, \forall j \in N_{\mathrm{b}} \tag{37}$$

$$\begin{cases} \mathrm{SoC}_{\mathrm{HRS},\min} \le \mathrm{SoC}_{\mathrm{HRS},j,t} \le \mathrm{SoC}_{\mathrm{HRS},\max} \\ \mathrm{SoC}_{\mathrm{HRS},j,t+1} = \mathrm{SoC}_{\mathrm{HRS},j,t} + \dfrac{H_{\mathrm{HRS},j,t} - D_{\mathrm{HRS},j,t}}{Q_{\mathrm{HRS},j}} \\ \qquad + \dfrac{\eta_{\mathrm{el}} P_{\mathrm{el},j,t} - P_{\mathrm{fc},j,t}/\eta_{\mathrm{fc}}}{Q_{\mathrm{HRS},j}} \Delta t \\ \mathrm{SoC}_{\mathrm{HRS},j,\mathrm{ini}} = \mathrm{SoC}_{\mathrm{HRS},j,\mathrm{fin}} \end{cases}, \forall j \in N_{\mathrm{b}} \tag{38}$$

where $d_{\mathrm{HV}m,t}$ is the hydrogen demand of m-th HV, $P_{\mathrm{el},j,t}$ is the power of electrolyzer at bus j and time t, $P_{\mathrm{fc},j,t}$ is the power of fuel cell at bus j and time t, $\mathrm{SoC}_{\mathrm{HRS},j,t}$ is the SoC of hydrogen storage at bus j and time t, $\mathrm{SoC}_{\mathrm{HRS},\min}$, $\mathrm{SoC}_{\mathrm{HRS},\max}$ denote the minimum and maximum allowable SoC limits of HRS, η_{el} is the electrolyzer efficiency for hydrogen production, η_{fc} is the fuel cell efficiency for electricity generation, $Q_{\mathrm{HRS},j}$ is the capacity of HRS at bus j, $\mathrm{SoC}_{\mathrm{HRS},j,\mathrm{ini}}$ and $\mathrm{SoC}_{\mathrm{HRS},j,\mathrm{fin}}$ represent the initial and final SoC values of HRS at bus j.

4. Stackelberg–Nash game framework

This section presents the two-level game framework that coordinates the centralized energy management of the AFV-integrated microgrid with decentralized station operations and user behavior. As shown in Fig. 2, the proposed framework employs a Stackelberg game structure at the system level and the Nash equilibrium game among stations at the lower level. The microgrid control center (leader) and V2G aggregator (follower) operate in a hierarchical relationship. The V2G aggregator serves as a central platform managing both EVs and HVs through their respective stations (i.e., CSs and HRSs).

4.1. Game formulation

The hierarchical game is characterized by the tuple:

$$\Gamma = \{I, S, U, H\} \tag{39}$$

in which:

(1) $I = \{0\} \cup I_{\mathrm{CS}} \cup I_{\mathrm{HRS}}$ denotes the set of players, with player 0 representing the microgrid operator, $I_{\mathrm{CS}} = \{1, 2, \ldots, N_{\mathrm{s}}\}$ the set of CS operators, and $I_{\mathrm{HRS}} = \{N_{\mathrm{s}+1}, \ldots, N_{\mathrm{s}} + N_{\mathrm{h}}\}$ the set of HRS operators.

(2) $S = S_0 \times \prod_{\{i \in I\mathrm{cs}\}} S_i \times \prod_{\{j \in I\mathrm{HRS}\}} S_j$ represents the joint strategy space, where:

$S_0 = \{(P_{\mathrm{grid},j,t}, P_{\mathrm{HRS},j,t}, P_{\mathrm{dc,EV}m,t}, P_{\mathrm{ch,EV}m,t})$: constraints (22)–(30), (33)–(38)$\}$ is the scheduling strategy space consisting for microgrid operator. $S_i = \{(c_{\mathrm{ch},i,t}, c_{\mathrm{V2G},i,t})$: constraints (9)–(21), (31), (32)$\}$ is the strategy space for CS operator i. $S_j = \{c_{\mathrm{HRS}i,t}$: constraints (9)–(21), (35)–(38)$\}$ is the strategy space for HRS operator j.

(3) $U = \{u_i: S \to \mathbb{R}\}_{\{i \in I\}}$ denotes the utility functions, where $u_0 = -C_{\mathrm{MG}}$ (microgrid), $u_{\mathrm{i}} = F_{\mathrm{CS},i}$ (CS), and $u_j = F_{\mathrm{HRS},j}$ (HRS) from eqs. (1), (6)–(7) respectively.

(4) H: $I \to 2\hat{}I$ defines the hierarchical structure, where $H(0) = I_{\mathrm{CS}} \cup I_{\mathrm{HRS}}$ indicates that the microgrid operator acts as the Stackelberg leader over all station operators, while $H(i) = \varnothing$ for all $i \in I_{\mathrm{CS}} \cup I_{\mathrm{HRS}}$ indicates that stations engage in simultaneous Nash competition at the lower level.

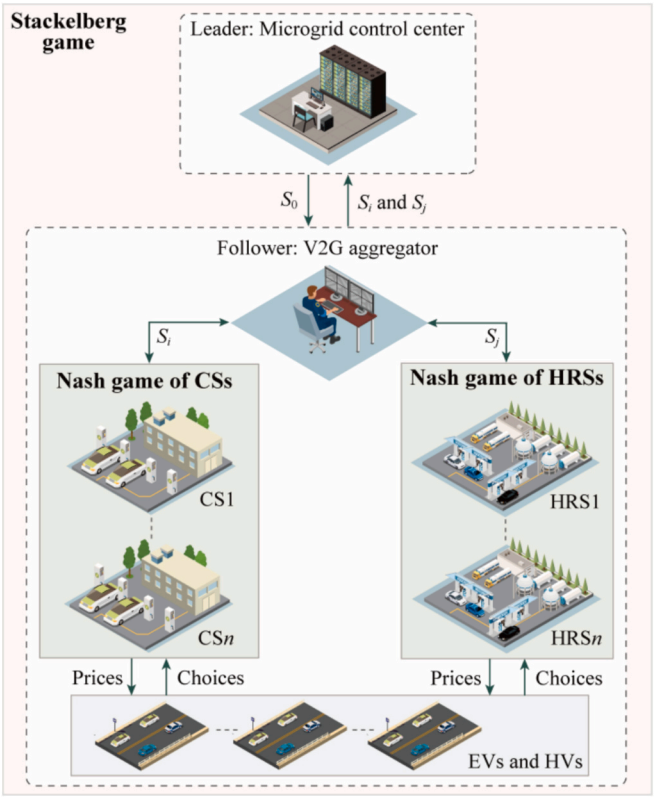


Fig. 2. The proposed Stackelberg–Nash game framework.

4.2. Stackelberg-nash game process

The proposed framework features a two-layer iterative process: an outer loop for Stackelberg equilibrium between the microgrid operator and station operators, and an inner loop for Nash equilibrium among station operators.

4.2.1. Step 1: System initialization

The game process commences with initializing the pricing strategy vectors over the entire time horizon. For all time periods $t \in T$, the pricing strategies of station operators are initialized as $\mathrm{x}^{(0)} = \{c^{(0)}_{\mathrm{ch},i,t}, c^{(0)}_{\mathrm{V2G},i,t}, c^{(0)}_{\mathrm{HRS}j,t}\}\{t = 1\}^T$ for all CS operators $i \in I_{\mathrm{CS}}$ and HRS operators $j \in I_{\mathrm{HRS}}$. Additionally, the data of EVs and HVs is generated based on the pricing strategies through the dynamic choice model defined in Section 3.2.

Algorithm 1. MESNEA.

Algorithm 1: MESNEA.

Initialization: $n \leftarrow 0, x^0 \in X, y^0 \in \Omega_{MG}$

while $||y^{(n+1)} - y^n|| > \varepsilon_y$ or $||x^{(n+1)} - x^n|| > \varepsilon_x$ do

Leader (Microgrid):

$$y^{(n+1)} = \arg\ \min \sum_{t \in T} C_{MG,t}(x^{(n)})$$

Extract: $\beta^{(n+1)} = \{\beta_{grid,t}{}^{(n+1)}, \beta_{HRS,t}{}^{(n+1)}\}_{t \in T}$

Followers (Nash Equilibrium):

$k \leftarrow 0, x^{(n,0)} \leftarrow x^n$

repeat

$\forall i \in I_{CS}$ (parallel):

$$\hat{x}_i^k = \arg \min_{x_i \in S_i} -F_{CS,i}(y^{(n+1)}, x_i, x_{-i}^{n,k})$$

$$x_i^{n,k+1} = x_i^{n,k} + \alpha_k(\hat{x}_i^k - x_i^{n,k})$$

$\forall j \in I_{HRS}$ (parallel):

$$\hat{x}_j^k = \arg \min_{x_j \in S_j} -F_{HRS,j}(y^{(n+1)}, x_j, x_{-j}^{n,k})$$

$$x_j^{n,k+1} = x_j^{n,k} + \alpha_k(\hat{x}_j^k - x_j^{n,k})$$

$k \leftarrow k + 1$

until $\left\|x^{(n,k+1)} - x^{(n,k)}\right\| \le \varepsilon_x$

$x^{(n+1)} \leftarrow x^{(n,k+1)}$

Update user behaviors:

$(p_{EVm,i,t}{}^{(n+1)}, p_{HVm,j,t}{}^{(n+1)}, \delta_{EVm,t}{}^{(n+1)}, \delta_{HVm,t}{}^{(n+1)}) = \Phi(x^{(n+1)})$

$n \leftarrow n + 1$

end

4.2.2. Step 2: Microgrid scheduling optimization

The microgrid operator determines the optimal scheduling strategy over the entire horizon by solving:

$$y^{(n+1)} = \arg\min \sum_{t \in T} C_{MG,t}\left(x^{(n)}\right) \tag{40}$$

where y = $\{P_{grid,j,t}, P_{HRS,j,t}, P_{dc,EVm,t}, P_{ch,EVm,t}\}$ represents the complete scheduling trajectory. The current EVs and HVs decisions is considered in this step from the previous iteration.

4.2.3. Step 3: Station competition with user response

(1) Step 3.1: Information broadcasting.

The microgrid operator announces the scheduling power to all CS operators and HRS operators through the V2G aggregator.

(2) Step 3.2: Nash competition.

Each CS operator solves:

$$x_i^* = \arg \min_{x_i \in S_i} -F_{CS,i}\left(y^{(n+1)}, x_i, x_{-i}\right), \forall i \in I_{CS} \tag{41}$$

where $x_i = \{c_{ch,i,t}, c_{V2G,i,t}\}$ represents the complete pricing trajectory of CS *i*.

Simultaneously, each HRS operator determines its optimal hydrogen price:

$$x_j^* = \arg \min_{x_j \in S_j} -F_{HRS,i}\left(y^{(n+1)}, x_j, x_{-j}\right), \forall j \in I_{HRS} \tag{42}$$

where $x_j = \{c_{HRSj,t}\}$ represents the complete pricing trajectory of HRS *i*. The Nash equilibrium emerges when no station can improve its total profit by unilaterally deviating from its pricing trajectory, considering the temporal interdependencies of user arrivals and service decisions.

4.2.4. Step 4: User behavior update

Based on the new equilibrium pricing strategies $x^{(n+1)}$ from Step 3, the EV and HV user behaviors are updated through the dynamic choice model.

4.2.5. Step 5: Convergence check and iteration

The iterative process continues until the Stackelberg-Nash equilibrium is achieved. Convergence is verified by examining the stability of scheduling decisions and pricing strategies:

$$\begin{cases} \left\|y^{(n+1)} - y^{(n)}\right\| \le \varepsilon_y \\ \left\|x^{(n+1)} - x^{(n)}\right\| \le \varepsilon_x \end{cases} \tag{43}$$

If convergence criteria are not satisfied, set $n = n + 1$ and return to Step 2. The converged equilibrium represents a stable state where the microgrid scheduling strategy can minimize total operation costs given the anticipated user behaviors and stations pricing strategies form a Nash equilibrium.

The two-layer structure ensures that the scheduling decisions of microgrid operator account for the decentralized pricing behavior of station operators, while station operators compete based on realistic user response patterns. This iterative process captures the complex interdependencies between energy dispatch, pricing strategies, and user behavior in the AFV-integrated microgrid system.

4.3. Solution method

The proposed bi-level optimization problem presents significant computational challenges due to its hierarchical structure and strategic interactions among multiple decision-makers. To address these challenges, this paper develops a MESNEA that integrates advanced optimization techniques for MILP with an accelerated iterative approach for Nash equilibrium computation. As shown in Algorithm 1, the MESNE alternates between solving the upper-level microgrid scheduling problem and computing the lower-level Nash equilibrium among stations.

Convergence is monitored through dual criteria: changes in microgrid operation decisions and stability of the Nash equilibrium. The iterative process continues until both thresholds are satisfied, ensuring a stable Stackelberg equilibrium.

The microgrid scheduling optimization constitutes a mixed-integer linear programming problem due to binary unit commitment variables, linear cost functions, and discrete vehicle-to-grid participation levels. The MESNEA employs commercial optimization solver (Gurobi) to handle this upper-level optimization problem.

The MESNEA employs a sophisticated dual-loop approach to compute the Nash equilibrium addressing both strategic interactions between operators and individual optimization complexity. The outer loop implements a momentum-based update mechanism that incorporates historical search directions to accelerate convergence and reduce oscillations common in game-theoretic algorithms. This approach provides faster convergence than traditional best-response dynamics and improved stability near equilibrium points. A diminishing step size rule is implemented to ensure guaranteed convergence while maintaining computational efficiency.

Each station operator solves an individual profit maximization problem to determine optimal pricing strategies given current strategies of competitors and updated microgrid pricing signals. Since demand functions follow logit choice models, these best response problems constitute standard nonlinear optimization problems. In MESNE, the gradient descent methods are used to solve the best response problem of each operator. Starting from the pricing strategy in the previous iteration, the algorithm iteratively updates pricing decisions until convergence. The gradient computation incorporates derivatives of expected demand functions with respect to pricing variables, which are readily available from the logit model structure.

5. Case study and discussion

To validate the effectiveness of the proposed two-level management framework for the AFV-integrated microgrid, several case studies are conducted in this section.

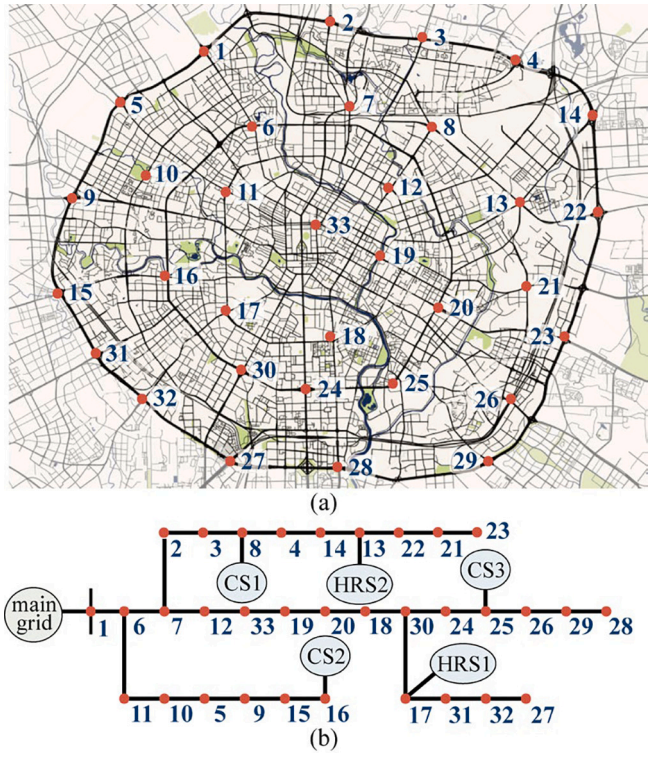


Fig. 3. Test platform developed based on Chengdu's CPTN, China [12]. (a) Transportation network. (b) Adapted 33-bus electrical grid.

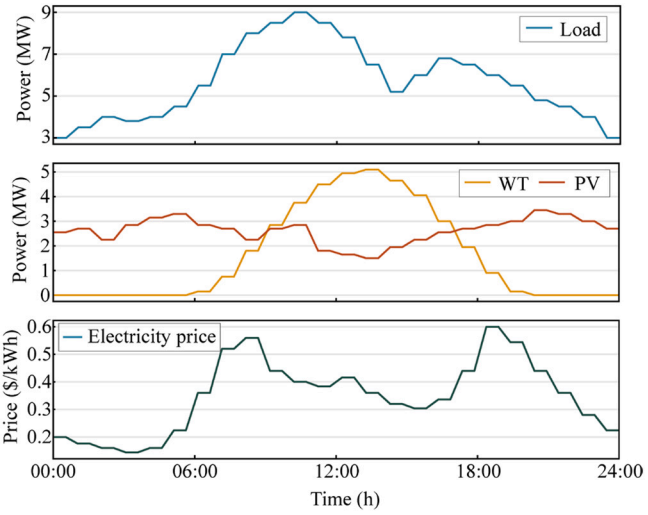


Fig. 4. The microgrid profiles for the test system.

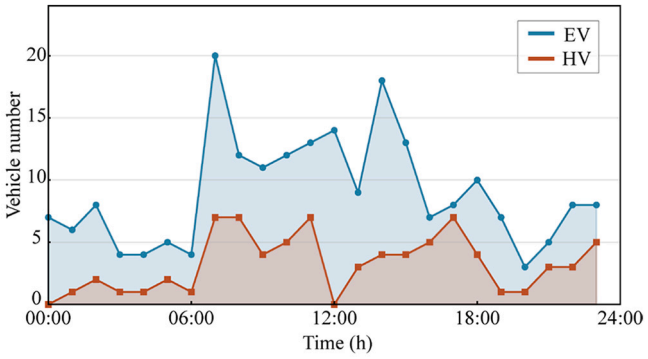


Fig. 5. The vehicles data for the test system.

Table 2
Main simulation parameters [2,24].

Parameters	Values	Parameters	Values
$P_{ch,EV,max}$	100 kW	$SoC_{HRS,min}$	20 %
$P_{dc,EV,max}$	100 kW	$SoC_{HRS,max}$	95 %
$SoC_{EV,max}$	95 %	η_{el}	0.02 kg/kWh
$SoC_{EV,min}$	10 %	η_{fc}	30 kWh/kg
Q_{EV}	100 kW/h	β_1	−1
$P_{el,17,max}$	2 MW	β_2	−0.2
$P_{el,13,max}$	1.5 MW	β_3	1
$P_{fc,17,max}$	2 MW	β_4	5
$P_{fc,13,max}$	1.5 MW	β_5	1
$Q_{HRS,17}$	600 kg	γ_3	5
$Q_{HRS,13}$	500 kg	γ_4	1

5.1. Test system

The test platform, illustrated in Fig. 3, comprises both the electrical distribution and Chengdu's transportation network. As shown in Fig. 3 (a), the transportation network covers 33 nodes distributed across the urban area of Chengdu. The electrical grid topology, depicted in Fig. 3 (b), connects these stations through a radial distribution network with the main grid connection at bus 1. In this work, three CSs are located at nodes 8, 16, and 25, respectively, and two HRSs are located at node 17 and node 13, respectively. Fig. 4 presents the 24-h operation profiles of the microgrid, including the load power, the RESs power and the time-

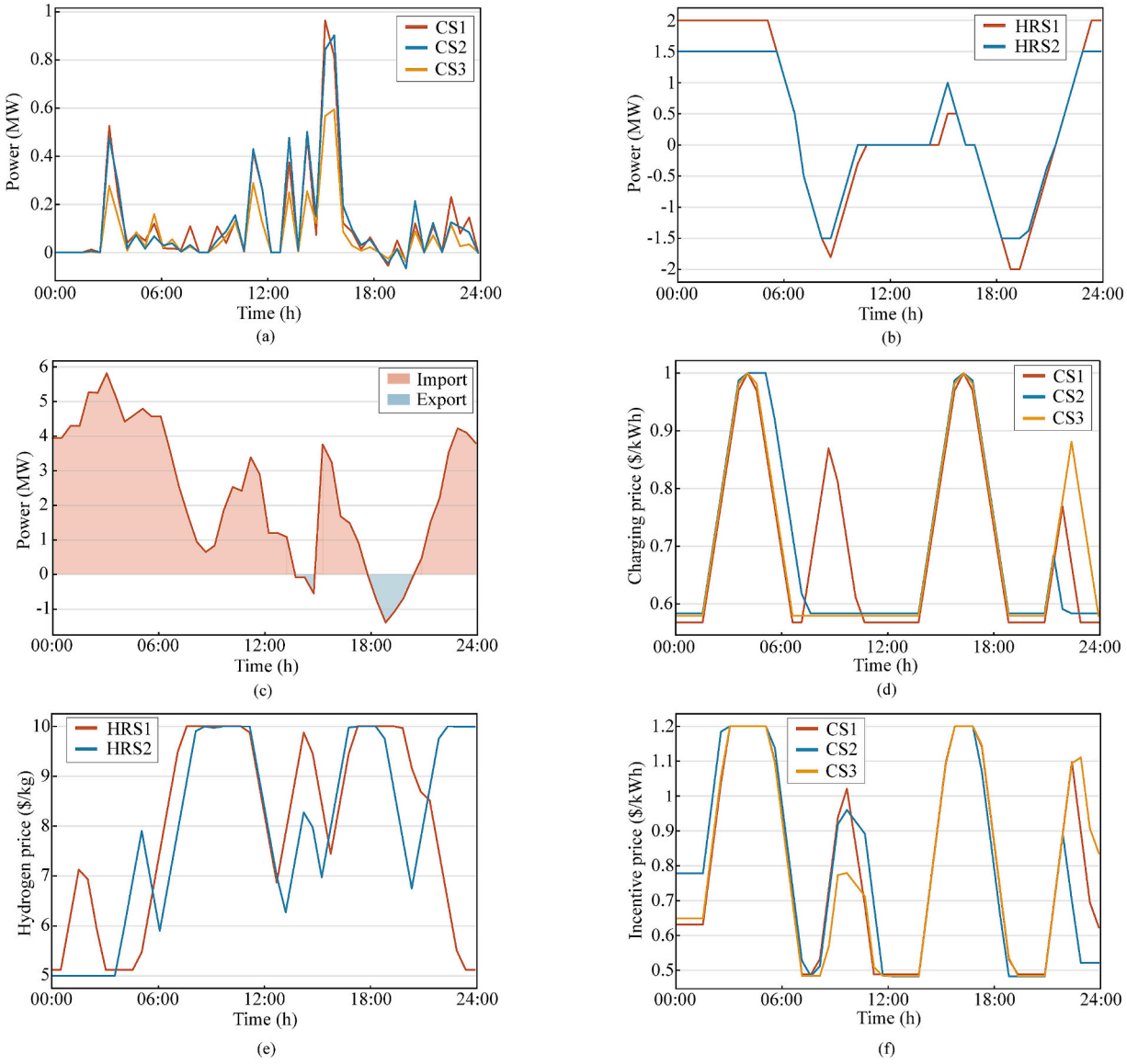


Fig. 6. Main simulation results of the proposed method. (a) Power of CSs. (b) Power of HRSs. (c) Power of main grid. (d) Charging price of CSs. (e) Hydrogen price of HRSs. (f) Incentive price of CSs.

of-use electricity price. The generation pattern of vehicle data, illustrated in Fig. 5, demonstrate realistic traffic flow characteristics with distinct peaks corresponding to commuting patterns. The total vehicle population during the 24-h simulation period comprises 216 EVs and 78 HVs. The simulation spans 24 h with a time step of $\Delta t = 0.5$ h. The main simulation parameters are shown in Table 2. The model is implemented in Python 3.11 on a laptop equipped with an Intel® Core™ Ultra 9185H processor and 32 GB of RAM.

To comprehensively evaluate the performance of the proposed framework, five case studies are designed:

(1) Case 1: The complete two-level framework is implemented with dynamic pricing optimization at the lower level and coordinated energy scheduling at the upper level. Both CSs and HRSs are managed optimally in the proposed framework.

(2) Case 2: The upper-level microgrid scheduling optimization is retained, but all stations are operated with predetermined fixed prices: \$0.7/kWh for EV charging, \$0.8/kWh for V2G incentives, and \$8.0/kg for hydrogen. This case evaluates the impact of dynamic pricing on system performance.

(3) Case 3: Only HRSs are managed optimally in the proposed framework, while CSs are operated with fixed pricing (\$0.7/kWh for charging, \$0.8/kWh for V2G). This case assesses the contribution of coordinated EV management to overall system efficiency.

(4) Case 4: CSs are managed optimally in the proposed framework, while HRSs are operated with a fixed price of \$8.0/kg. This scenario isolates the impact of hydrogen system coordination on microgrid operations.

(5) Case 5: Neither CSs nor HRSs participate in the coordinated optimization. All stations are operated with fixed prices.

These cases are designed to systematically evaluate the contributions of different system components and coordination mechanisms.

5.2. Simulation results of the proposed method

This section presents the simulation results of Case 1, where the proposed two-level management framework is fully implemented. Fig. 6 illustrates the 24-h operation profiles of the AFV-integrated microgrid under the proposed optimization framework.

As shown in Fig. 6(a), the power profiles of CSs demonstrate strategic load management throughout the day. The power consumption peaks

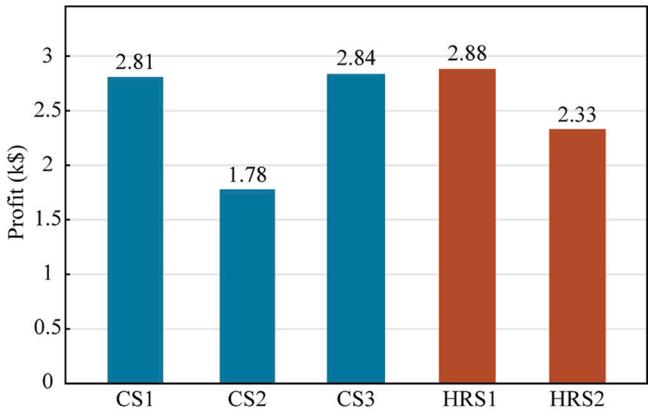


Fig. 7. Profit for all CSs and HRSs.

during midday hours (14:00–16:00), reaching approximately 0.9 MW for CS1, coinciding with high PV generation and moderate electricity prices. This pattern indicates effective utilization of RESs for EV charging. Notably, during evening peak hours (18:00–20:00), All stations have implemented V2G functionality, demonstrating price response behavior.

The operation power of HRSs is depicted in Fig. 6(b). Both HRS1 and HRS2 are operated in hydrogen production mode (positive power) during the initial hours (00:00–08:00) when electricity prices are low. A significant operation shift occurs during peak price periods, where both stations switch to power generation mode (negative power) during the evening peak (18:00–20:00). This bidirectional operation demonstrates the value of hydrogen systems in providing grid flexibility and peak shaving capabilities.

Fig. 6(c) presents the net grid power exchange profile, clearly illustrating the microgrid's strategic interaction with the main grid. The microgrid imports power (positive values) during off-peak hours. Conversely, during midday hours when renewable generation power is high, the microgrid reduced imports. In the early evening hours, the exported power (negative values) increases due to higher prices, indicating effective local resource utilization through V2G and hydrogen-based generation.

The optimal pricing profiles are shown in Figs. 6(d)-(f). The charging price trajectories in Fig. 6(d) exhibit pronounced time-varying characteristics aligned with system conditions. Hydrogen pricing strategies, illustrated in Fig. 6(e), show the variations ranging from $5.0/kg to $10.0/kg. The price volatility reflects the dynamic balance between hydrogen production costs, storage levels, and demand patterns. The V2G incentive pricing is shown in Fig. 6(f). This incentive structure, exceeding the concurrent charging prices, ensures profitable V2G participation while providing valuable grid services during critical periods.

Fig. 7 presents the economic outcomes for all stations under the proposed framework, demonstrating the financial viability of the coordinated optimization approach. CS1 and CS3 achieve similar profit levels of $2810 and $2840 respectively, benefiting from their strategic positions at major traffic intersections with high vehicle flow. In contrast, CS2 shows significantly lower profitability at $1780. This disparity can be attributed to multiple factors revealed through our network analysis: (1) CS2 is located on a secondary route with limited connectivity, resulting in substantially lower traffic volume compared to CS1 and CS3. (2) Origin-destination analysis indicates that CS2 falls outside the primary commuting corridors, with most EV trips originating from residential areas closer to CS1 and terminating near commercial centers adjacent to CS3. (3) CS2's intermediate position between CS1 and CS3 creates a competitive disadvantage, as users often choose neighboring stations when prices are similar, leading to customer diversion. (4) The combination of unfavorable location and traffic patterns amplifies the impact of any service quality differences on station selection. The hydrogen refueling stations demonstrate robust profitability, with HRS1 earning $2880 and HRS2 generating $2330 in daily profits.

Table 3
Performance comparison across different cases.

	Case 1	Case 2	Case 3	Case 4	Case 5
Microgrid cost (k$)	18.93	20.92	29.67	26.73	42.29
Total CS profit (k$)	7.42	6.13	4.31	6.85	4.31
Total HRS profit (k$)	5.21	4.83	4.21	2.07	2.07
EV participation rate	93.6 %	90.1 %	90.1 %	92.8 %	90.1 %
HV participation rate	87.1 %	82.8 %	84.3 %	82.8 %	82.8 %

5.3. Case comparison analysis

Table 3 presents the performance metrics across all five cases. The lowest microgrid cost of $18,930 appears in case 1, while costs progressively increase with reduced coordination: case 2 ($20,920, +10.5 %), case 4 ($26,730, +41.2 %), case 3 ($29,670, +56.7 %), and case 5 ($42,290, +123.4 %). It can be seen that CS coordination has greater impact on cost reduction than HRS coordination, as evidenced by the $2940 difference between cases 3 and 4. This is due to the fact that uncoordinated EV charging increases the cost of power purchases from the main grid. The dramatic cost escalation in case 5 underscores the critical importance of AFV-microgrid integration, where uncoordinated operations more than double operation expenses due to inefficient peak load management. The comparison between cases 1 and 2 highlights the value of dynamic pricing optimization: compared to fixed pricing, the proposed method reduces costs by $1990 (9.5 %) while increasing total station profit from $10,960 to $12,630 (+15.2 %). This shows that dynamic pricing reduces system costs while improving profits for both CSs and HRSs.

Station profits vary significantly across cases, with CS profits ranging from $4310 (cases 3 and 5) to $7420 (case 1), and HRS profits spanning from $2070 (cases 4 and 5) to $5210 (case 1). Vehicle participation rates show that the proposed method achieves the highest values (93.6 % for EVs, 87.1 % for HVs), while other cases maintain relatively stable rates around 90.1 % for EVs and 82.8 % for HVs, with case 3 showing intermediate HV participation at 84.3 %. These results demonstrate that the proposed hierarchical framework can effectively optimize all three key performance dimensions: achieving 55.2 % microgrid cost reduction, nearly doubling total station profits, and improving vehicle participation rates compared to uncoordinated operations, thereby validating the superiority of integrated AFV-microgrid management in smart city environments.

5.4. Sensitivity analysis

To evaluate the robustness of the proposed framework under varying user behavior characteristics, a sensitivity analysis is conducted by examining the impact of key behavioral parameters on system performance. Fig. 8 presents the sensitivity analysis results across different combinations of price coefficient β_1 and distance coefficient β_2 in the dynamic choice model, ranging from −2.0 to −0.1 and − 0.5 to −0.05, respectively. In this work, $\gamma_1 = 0.1^*\beta_1$ and $\gamma_2 = \beta_2$.

Fig. 8(a) demonstrates that microgrid operation costs exhibit strong sensitivity to user behavioral parameters. When users are highly sensitive to both price and distance ($\beta_1 = -2.0$, $\beta_2 = -0.5$), the system achieves minimal operation costs of $13,530. As sensitivity decreases (coefficients approach zero), costs progressively increase, reaching $20,380 when both sensitivities are low ($\beta_1 = -0.1$, $\beta_2 = -0.05$). This 50.6 % cost increase is primarily driven by the higher EV charging and HV refueling demand, which increases electricity procurement and operation costs for the microgrid.

The total station profit patterns (CS + HRS) shown in Fig. 8(b) reveal

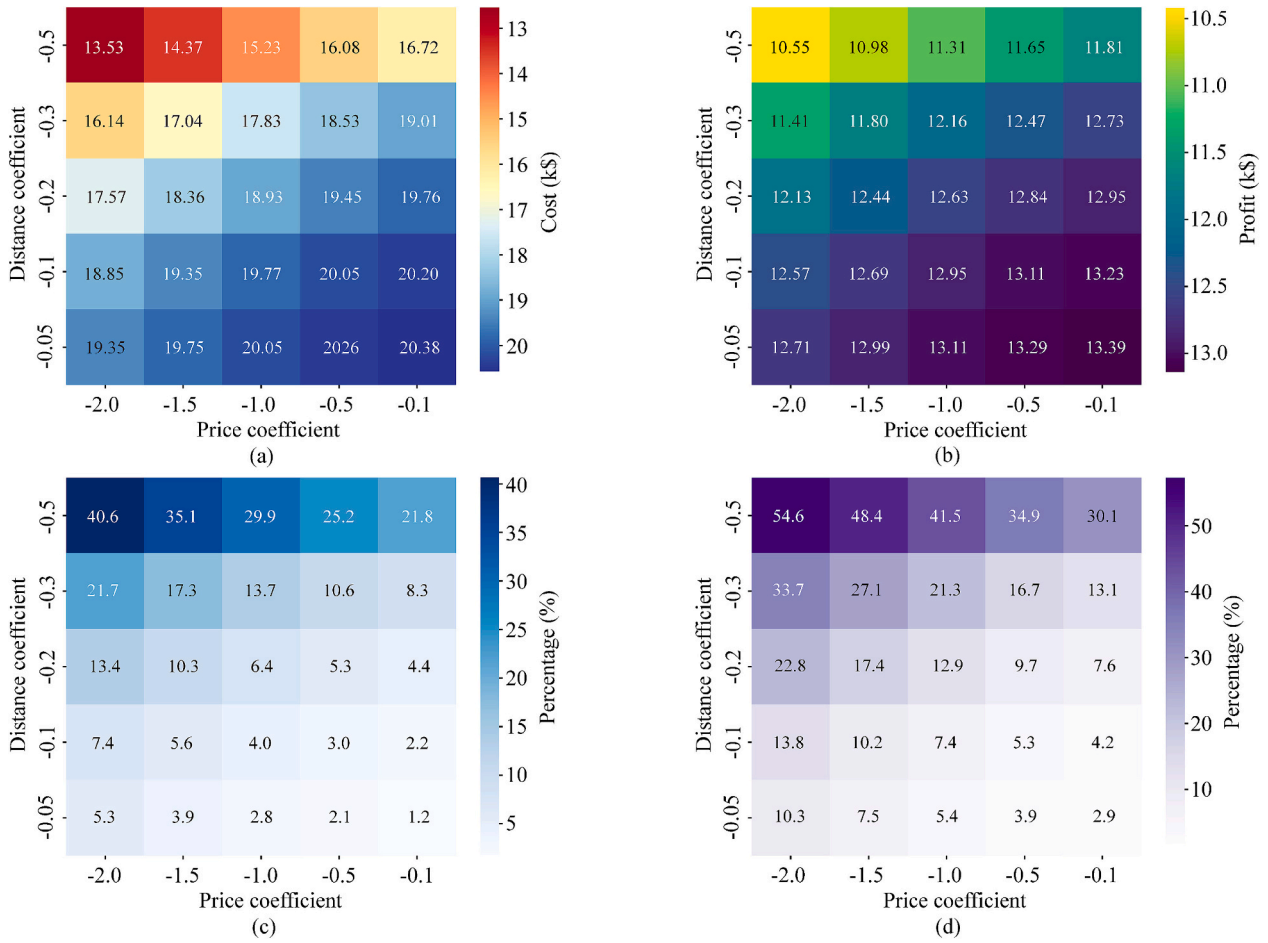


Fig. 8. (a) Microgrid cost. (b) Total stations profit. (c) Ratio of EVs without charging. (d) Ratio of HVs without refueling.

an inverse relationship with user sensitivity. Maximum profitability of \$13.39 k occurs under low sensitivity conditions ($\beta_1 = -0.1$, $\beta_2 = -0.05$), while profits decrease to \$10.55 k under high sensitivity conditions. This pattern directly reflects the impact of service demand—when user sensitivity is low, more vehicles utilize the charging and refueling services, generating higher profits for the stations.

Fig. 8(c) shows that the percentage of EVs choosing not to charge varies from 1.2 % under high sensitivity conditions ($\beta_1 = -2.0$, $\beta_2 = -0.5$) to 40.6 % when both sensitivities are low. Similarly, Fig. 8(d) demonstrates that HV non-refueling rates range from 2.9 % to 54.6 %. This substantial variation indicates that effective price signals and convenient station placement are critical for maintaining high service utilization rates.

The sensitivity analysis reveals fundamental trade-offs in the performance of the proposed framework. These findings validate that the proposed two-level framework can achieve robust performance across diverse behavioral scenarios while highlighting the importance of understanding local user characteristics for successful implementation.

6. Conclusion

This paper proposes a Stackelberg-Nash game framework to enhance the management of AFV-integrated microgrid in urban CPTN. The proposed framework employs a two-level structure where the microgrid operator acts as the leader and the V2G aggregator as the follower, with Nash equilibrium games modeling competition among CSs and among HRSs. This framework integrates an optimal scheduling model, supported by a dynamic choice model, to optimize energy scheduling and a game-theoretic pricing and incentive mechanism to boost CS and HRS profitability through V2G participation.

Case studies demonstrate the framework's effectiveness, with dynamic pricing and tailored strategies proving advantageous across varying operational scenarios. The proposed method can ensure efficient coordination between centralized energy management and decentralized station competition, providing an effective solution for integrating large-scale AFV fleets into urban microgrid operations while optimizing energy distribution in the microgrid. Future research could explore real-time adaptations and emerging technologies to further enhance its applicability.

CRediT authorship contribution statement

Yuyang Wan: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ning Wang:** Writing – review & editing, Writing – original draft, Validation, Software, Funding acquisition, Formal analysis, Conceptualization. **Ershun Du:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition. **Xueshan Liu:** Writing – review & editing, Visualization, Methodology, Formal analysis. **Yanbo Wang:** Writing – original draft, Supervision, Project administration, Formal analysis. **Zhe Chen:** Writing – review & editing, Supervision, Project administration, Methodology. **Chongqing Kang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the Carbon Neutrality and Energy System Transformation project, Tsinghua University.

Data availability

Data will be made available on request.

References

[1] Chatterjee S, Chaudhuri R, Vrontis D, Bresciani S. Exploring the effect of government incentives on electric vehicle purchase intention in smart cities. J Clean Prod 2024;477:143841. https://doi.org/10.1016/j.jclepro.2024.143841.
[2] Yan J, Cheng Y, Zhang F, Li M, Zhou N, Jin B, et al. Research on multimodal techniques for arc detection in railway systems with limited data. Struct Health Monit 2025. https://doi.org/10.1177/14759217251336797. 14759217251336797.
[3] Wang N, Jiang Y, Wang Y, Mou D, Liu Y, Hu W, et al. Advanced operation and control schemes of retired batteries in DC microgrid for secondary usage. IEEE Trans Smart Grid 2025;16:3155–66. https://doi.org/10.1109/TSG.2025.3555181.
[4] Wan Y, Wang Y, Yu H, Zhang H, Chen Z. Smart mobile power bank: a novel grid-friendly mobile microgrid for power grid with high penetration of renewable vehicles. IEEE Trans Transp Electrification 2025;11:2287–98. https://doi.org/10.1109/TTE.2024.3420399.
[5] Navinkumar TM, Bharatiraja C. Sustainable hydrogen energy fuel cell electric vehicles: a critical review of system components and innovative development recommendations. Renew Sust Energ Rev 2025;215:115601. https://doi.org/10.1016/j.rser.2025.115601.
[6] Ahmed I, Maaruf M, Ali A, AlMuhaini M, Khalid M. A review on challenges and future of green charging networks for zero emission electric transportation. Energy Convers Manag X 2025;25:100867. https://doi.org/10.1016/j.ecmx.2024.100867.
[7] Qi W, Yang J, Zhang Z, Wu J, Lan P, Xiang S. Investigation on thermal management of cylindrical lithium-ion batteries based on interwound cooling belt structure. Energy Convers Manag 2025;340:119962. https://doi.org/10.1016/j.enconman.2025.119962.
[8] Guan A, Zhou S, Gu W, Wu Z, Ai X, Fang J, et al. A dynamic model-based minute-level optimal operation strategy for alkaline Electrolyzers in wind-hydrogen systems. IEEE Trans Sustain Energy 2025;16:2157–70. https://doi.org/10.1109/TSTE.2025.3548052.
[9] Ren F, Lin X, Wei Z, Zhai X, Yang J. A novel planning method for design and dispatch of hybrid energy systems. Appl Energy 2022;321:119335. https://doi.org/10.1016/j.apenergy.2022.119335.
[10] Hussain A, Kazemi N, Musilek P. Clustering-based EV suitability analysis for grid support services. Energy 2025;320:134970. https://doi.org/10.1016/j.energy.2025.134970.
[11] Li D, Leung M-F, Tang J, Wang Y, Hu J, Wang S. Generative self-supervised learning for cyberattack-resilient EV charging demand forecasting. IEEE Trans Intell Transp Syst 2025:1–10. https://doi.org/10.1109/TITS.2025.3545577.
[12] Upputuri RP, Subudhi B. A comprehensive review and performance evaluation of bidirectional charger topologies for V2G/G2V operations in EV applications. IEEE Trans Transp Electrification 2024;10:583–95. https://doi.org/10.1109/TTE.2023.3289965.
[13] Agredano-Torres M, Xu Q. Decentralized power management of hybrid hydrogen electrolyzer—supercapacitor systems for frequency regulation of low-inertia grids. IEEE Trans Ind Electron 2025;72:8072–81. https://doi.org/10.1109/TIE.2025.3528468.
[14] Wan Y, Hu Y, Liu X, Zhou Q, Wang N, Wang Y, et al. Routing and scheduling of smart mobile power banks for mobile charging and vehicle-to-grid services. IEEE Trans Transp Electrification 2025;11:8054–64. https://doi.org/10.1109/TTE.2025.3535500.
[15] Qi W, Lan P, Yang J, Chen Y, Zhang Y, Wang G, et al. Multi-U-style micro-channel in liquid cooling plate for thermal management of power batteries. Appl Therm Eng 2024;256:123984. https://doi.org/10.1016/j.applthermaleng.2024.123984.
[16] Maaruf M, El-Ferik S. Neural learning fault-tolerant control of fuel cell–battery–ultracapacitor-based hybrid electric vehicle. J Energy Storage 2024;98:112892. https://doi.org/10.1016/j.est.2024.112892.
[17] Lehtola T. Vehicle-to-grid applications and battery cycle aging: a review. Renew Sust Energ Rev 2025;208:115013. https://doi.org/10.1016/j.rser.2024.115013.
[18] Xu L, Dou C, Yue D, Zhang Y, Zhang B, Li H, et al. Hybrid modeling and switching control of electric vehicle aggregation for frequency regulation. IEEE Trans Sustain Energy 2025;16:1860–73. https://doi.org/10.1109/TSTE.2025.3540253.
[19] Kiani S, Sheshyekani K, Dagdougui H. An extended state space model for aggregation of large-scale EVs considering fast charging. IEEE Trans Transp Electrification 2023;9:1238–51. https://doi.org/10.1109/TTE.2022.3179311.
[20] Yin W, Jia L, Ji J. Energy optimal scheduling strategy considering V2G characteristics of electric vehicle. Energy 2024;294:130967. https://doi.org/10.1016/j.energy.2024.130967.
[21] Irfan M, Deilami S, Huang S, Tahir T, Veettil BP. Optimizing load frequency control in microgrid with vehicle-to-grid integration in Australia: based on an enhanced control approach. Appl Energy 2024;366:123317. https://doi.org/10.1016/j.apenergy.2024.123317.
[22] Liang Z, Qian T, Korkali M, Glatt R, Hu Q. A vehicle-to-grid planning framework incorporating electric vehicle user equilibrium and distribution network flexibility enhancement. Appl Energy 2024;376:124231. https://doi.org/10.1016/j.apenergy.2024.124231.
[23] Xie S, Xie L, Wu Q, Shu S, Chen Y, Yang Q, et al. Charging pricing in power-traffic systems with price-elastic demand: a quasi-variational inequality approach. IEEE Trans Power Syst 2025:1–13. https://doi.org/10.1109/TPWRS.2025.3566413.
[24] Wan Y, Wang N, Liu X, Wang Y, Blaabjerg F, Chen Z. Inertia-emulation-based fast frequency response from EVs: a multi-level framework with game-theoretic incentives and DRL. IEEE Trans Smart Grid 2025:1. https://doi.org/10.1109/TSG.2025.3603256.
[25] Wang M, Mu Y, Li F, Jia H, Li X, Shi Q, et al. State space model of aggregated electric vehicles for frequency regulation. IEEE Trans Smart Grid 2020;11:981–94. https://doi.org/10.1109/TSG.2019.2929052.
[26] Qiu D, Wang Y, Sun M, Strbac G. Multi-service provision for electric vehicles in power-transportation networks towards a low-carbon transition: a hierarchical and hybrid multi-agent reinforcement learning approach. Appl Energy 2022;313:118790. https://doi.org/10.1016/j.apenergy.2022.118790.
[27] Wan Y, Zhang H, Hu Y, Wang Y, Liu X, Zhou Q, et al. A novel energy management framework for retired battery-integrated microgrid with peak shaving and frequency regulation. Energy 2024;313:133907. https://doi.org/10.1016/j.energy.2024.133907.
[28] Qiao W, Han Y, Si F, Wang J, Zhao Q. A carbon-tax-based pricing scheme for vehicle scheduling in coupled power-traffic networks. IEEE Trans Transp Electrification 2024;10:4029–41. https://doi.org/10.1109/TTE.2023.3313125.
[29] Qian T, Liang Z, Chen S, Hu Q, Wu Z. A tri-level demand response framework for EVCS flexibility enhancement in coupled power and transportation networks. IEEE Trans Smart Grid 2025;16:598–611. https://doi.org/10.1109/TSG.2024.3417294.
[30] Kumar P, Channi HK, Kumar R, Rajiv A, Kumari B, Singh G, et al. A comprehensive review of vehicle-to-grid integration in electric vehicles: powering the future. Energy Convers Manag X 2025;25:100864. https://doi.org/10.1016/j.ecmx.2024.100864.
[31] Bayrak G, Başaran K, Lazaroiu AC. Integrating hydrogen-powered fuel cell electric buses into grid-forming microgrids: a solution for emergency energy needs. Renew Energy 2025;250:123363. https://doi.org/10.1016/j.renene.2025.123363.
[32] Lai X, Yang J, Wen F, Yang Dong Z. Operational economics of renewable energy-hydrogen system with hydrogen-powered transportation. IEEE Trans Smart Grid 2025;16:718–27. https://doi.org/10.1109/TSG.2024.3465505.
[33] Ali A, Shaaban MF, Mahmoud K. Optimizing hydrogen systems and demand response for enhanced integration of RES and EVs in smart grids. IEEE Trans Smart Grid 2025;16:1366–78. https://doi.org/10.1109/TSG.2024.3491774.
[34] Ghirardi E, Brumana G, Franchini G, Aristolao N, Vedovati G. The role of hydrogen storage and electric vehicles in grid-isolated hybrid energy system with high penetration of renewable. Energy Convers Manag 2024;302:118154. https://doi.org/10.1016/j.enconman.2024.118154.
[35] Federici L, Tribioli L, Cozzolino R, Mendecka B, De Santis M. Investigation on battery and fuel cell electric vehicle-to-grid potential for microgrid frequency regulation. Int J Hydrog. Energy 2025. https://doi.org/10.1016/j.ijhydene.2025.03.085. S0360319925011759.
[36] Rezaee Jordehi A, Mansouri SA, Tostado-Véliz M, Hakimi SM, Safaraliev M, Nasir M. A three-level model for integration of smart homes, electric vehicle charging stations and hydrogen fuelling stations in reconfigurable microgrids considering vehicle-to-infrastructure (V2I) technology. Energy 2025;314:134330. https://doi.org/10.1016/j.energy.2024.134330.
[37] Wang Q, Wang Y, Chen Z. Stackelberg-equilibrium-based collaborative charging management strategy for renewable fuel vehicles in regional integrated electricity-hydrogen system. Appl Energy 2025;377:124617. https://doi.org/10.1016/j.apenergy.2024.124617.
[38] Singh M, Singla MK, Beryozkina S, Gupta J, Safaraliev M. Hydrogen vehicles and hydrogen as a fuel for vehicles: A-state-of-the-art review. Int J Hydrog Energy 2024;64:1001–10. https://doi.org/10.1016/j.ijhydene.2024.03.325.
[39] Zhang Y, Xu S, Song Y, Qi W, Guo Q, Li X, et al. Real-time global optimal energy management strategy for connected PHEVs based on traffic flow information. IEEE Trans Intell Transp Syst 2024;25:20032–42. https://doi.org/10.1109/TITS.2024.3464757.
[40] Chang X, Ma T, Wu R. Impact of urban development on residents' public transportation travel energy consumption in China: an analysis of hydrogen fuel cell vehicles alternatives. Int J Hydrog Energy 2019;44:16015–27. https://doi.org/10.1016/j.ijhydene.2018.09.099.
[41] Tao Y, Qiu J, Lai S, Zhang X, Wang G. Collaborative planning for electricity distribution network and transportation system considering hydrogen fuel cell

vehicles. IEEE Trans Transp Electrification 2020;6:1211–25. https://doi.org/10.1109/TTE.2020.2996755.
[42] Wang Y, Liu J, Qu Z. Multi-stage collaborative planning of electricity-hydrogen-transportation coupling network considering carbon emission reduction. Electr Power Syst Res 2024;228:110071. https://doi.org/10.1016/j.epsr.2023.110071.
[43] Zhang W, Su J, Li Y. Flexible scheduling of the electric–hydrogen coupling system based on EV–HFEV mixed traffic flow. IEEE Trans Transp Electrification 2024;10:5921–35. https://doi.org/10.1109/TTE.2023.3323354.
[44] McFadden D. Conditional logit analysis of qualitative choice behavior. 1972.
[45] Farivar M, Low SH. Branch flow model: relaxations and convexification—part I. IEEE Trans Power Syst 2013;28:2554–64. https://doi.org/10.1109/TPWRS.2013.2255317.